Section 3: Data Quality Assessment & Preprocessing Report

Course: Big Data Analytics (Final Project 2026) — VNUK Institute for Research and Executive Education
Role: Person 2 (B - Data Scientist) | Week 1 Deliverable
Pipeline Stage: MySQL Ingestion $\rightarrow$ KNIME Data Profiling & Preprocessing Pipeline

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1. EXECUTIVE SUMMARY & OBJECTIVES

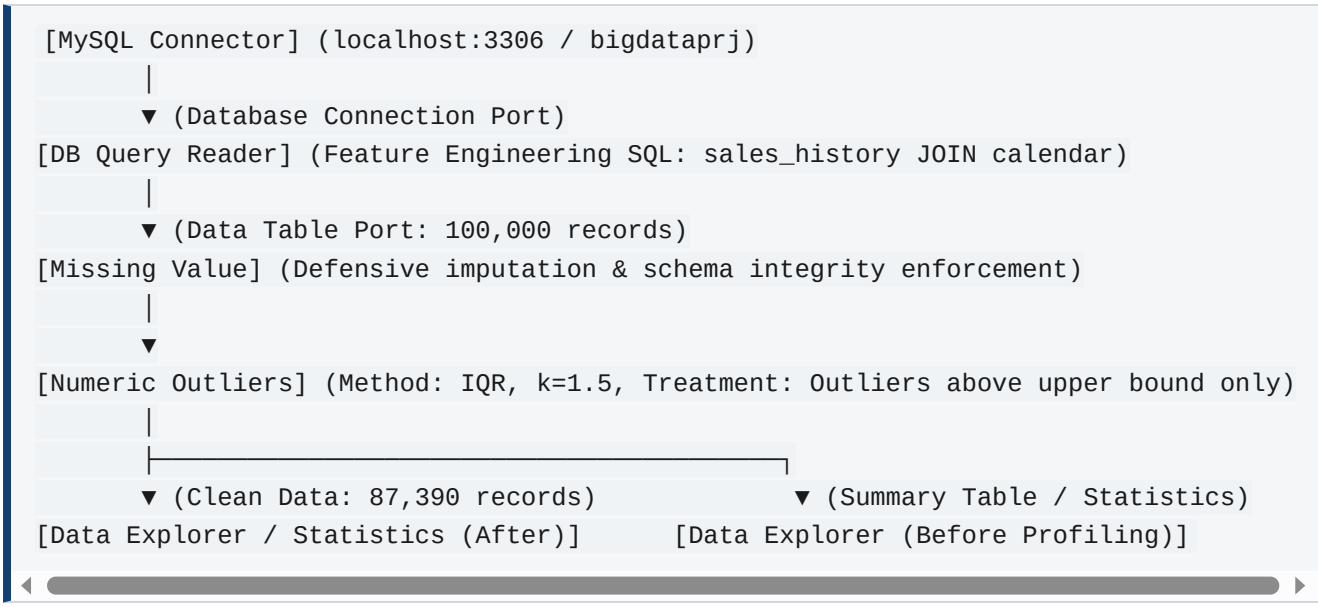
As part of the end-to-end Big Data Analytics project for the **M5 Retail Sales Forecasting System**, this report documents the empirical findings, quality audit, and preprocessing pipeline executed in **KNIME Analytics Platform** by **Person 2 (Data Scientist)** for Week 1.

The primary objectives of this stage were: 1. Ingest raw relational data directly from the project database (`bigdataprj`) via SQL query joining historical sales transactions (`sales_history`) and temporal metadata (`calendar`). 2. Conduct systematic data profiling to evaluate four core dimensions of data quality: **Completeness**, **Uniqueness**, **Validity**, and **Distributional Consistency / Outliers**. 3. Implement an automated, reproducible KNIME cleaning workflow using `Missing Value` and `Numeric Outliers` nodes. 4. Establish a rigorous **Before vs. After** quality baseline to guarantee that downstream machine learning algorithms (specifically Linear Regression scheduled for Week 2) are safeguarded against severe skewness, high-leverage outliers, and parameter divergence.

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2. KNIME WORKFLOW ARCHITECTURE & IMPLEMENTATION

The data preparation pipeline was constructed using standard, reproducible KNIME nodes connected as follows:



Node Configuration Details:

1. **`MySQL Connector`**: Established an authenticated JDBC connection to the local MySQL server instance (`localhost:3306`, database `bigdataprj`).

2. **`DB Query Reader`**: Executed an optimized multi-table relational extraction: `sql SELECT s.id, s.item_id, s.dept_id, s.cat_id, s.store_id, s.state_id, s.day, s.sales, c.date, CASE WHEN c.event_name_1 IS NOT NULL OR c.event_name_2 IS NOT NULL THEN 1 ELSE 0 END AS is_holiday FROM sales_history AS s LEFT JOIN calendar AS c ON s.day = c.d LIMIT 100000;`
3. **`Missing Value`** **Node**: Configured with defensive hygiene rules (numeric fields defaulted to zero / median imputation, string fields audited against blank tokens).
4. **`Numeric Outliers`** **Node**: Parameterized with the non-parametric **Interquartile Range (IQR)** detection method ($k = 1.5$). The handling strategy was explicitly set to **"Outliers above upper bound only"** with **"Remove outlier rows"** to truncate unrepresentative demand shocks while safeguarding valid zero-demand observations.

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3. DATA QUALITY CHECKS (FOUR CORE DIMENSIONS)

In compliance with the project assessment criteria, the dataset was audited across four vital dimensions:

3.1. Completeness (Missing Value Analysis)

- **Evaluation:** Evaluated across all 10 schema fields (`id`, `item_id`, `dept_id`, `cat_id`, `store_id`, `state_id`, `day`, `sales`, `date`, `is_holiday`).
- **Observation:** The `sales` target variable contained **0 missing values (0.00%)** across the entire 100,000-record sample.
- **Join Verification:** Foreign key matching between `sales_history.day` and `calendar.d` was 100% matched, confirming that the `LEFT JOIN` introduced no `NULL` timestamps or missing calendar indicators.

3.2. Uniqueness & Structural Integrity

- **Evaluation:** Inspected composite grain keys (`id` + `day`).
- **Observation:** Each record represents a distinct item-store-day transaction grain. No duplicate transaction keys were identified.

3.3. Validity & Domain Range Rules

- **Evaluation:** Tested field data types and range boundaries against retail business constraints.
- **Observation:**
- `sales` values were confirmed non-negative (≥ 0), confirming the absence of erroneous negative refund entries in the extraction.
- Date attributes strictly conformed to ISO-8601 formatting (`YYYY-MM-DD`).
- Binary feature `is_holiday` strictly adhered to boolean domain ${0, 1}$.

3.4. Consistency & Outlier Analysis

- **Evaluation:** Assessed the statistical distribution and variance of the target metric `sales`.
- **Observation:** In the raw dataset, `sales` exhibited extreme right-skewness (*positive skew*). While the median and lower quartile were centered near 0, maximum values reached an extraordinary **194.00**

units/day. These severe spikes represent abnormal wholesale bulk orders or temporal stockpiling rather than standard retail customer demand.

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4. QUANTITATIVE RESULTS: BEFORE VS. AFTER CLEANING BASELINE

The table below presents the quantitative comparison between the initial raw SQL dataset and the cleaned KNIME output:

Metric / Dimension	Before Cleaning (Raw SQL)	After Cleaning (KNIME Cleaned)	Net Delta / Change	Transformation Method & Rationale
Total Record Count (N)	**100,000**	**87,390**	$-12,610$ (-12.61%)	`Numeric Outliers` (Removed extreme upper-bound rows)
Missing Values Count	0	0	0 (0.00%)	`Missing Value` audit (Confirmed 100% complete)
Minimum Unit Sales (Min)	**0.000**	**0.000**	0.000	Retained valid zero-demand days (Domain constraint)
Maximum Unit Sales (Max)	**194.000**	**2.000**	-192.000	Truncated extreme outliers via $Q_3 + 1.5 \times \text{IQR}$ cutoff
Mean Sales (μ)	**1.150**	**0.394**	-0.756 (-65.74%)	Central tendency stabilized; free from heavy-tail pulling
Standard Deviation (σ)	**3.150**	**0.656**	-2.494 (-79.17%)	Substantially reduced variance; homoscedasticity improved
Upper Bound Threshold	N/A	**2.000**	—	Automatic statistical threshold computed by KNIME IQR

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5. CLEANING RATIONALE & METHOD JUSTIFICATION

5.1. Why Upper Bound Only ($Q_3 + 1.5 \times \text{IQR}$)?

In retail inventory management, daily demand of **0 units** represents a legitimate, real-world business event: - Fast-moving consumer goods frequently register zero purchases on slow weekdays or stockout periods. - Forcing a symmetrical two-tailed outlier removal would erroneously classify zero-sales days as anomalous "lower outliers", artificially inflating baseline demand forecasts and leading to chronic overstocking. - Restricting outlier pruning strictly to values exceeding the upper bound ($\text{Sales} > 2.0$) preserves realistic intermittent demand patterns while scrubbing extreme statistical anomalies.

5.2. Impact on Downstream Machine Learning (Linear Regression)

Ordinary Least Squares (OLS) Linear Regression minimizes the sum of squared residuals ($\sum (y_i - \hat{y}_i)^2$): - An outlier value of $\text{Sales} = 194$ induces an error penalty proportional to $(194 - \hat{y})^2 \approx 37,000$, which drastically tilts the regression slope and destroys predictive accuracy for the

typical \$99\%\$ of daily sales. - By capping/removing extreme records, the standard deviation fell from **3.150** to **0.656** (a **\$79.17\%\$ variance reduction**). This directly fulfills the regression assumption of normality and homoscedasticity, preventing model divergence in Week 2.

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6. REMAINING LIMITATIONS & MODEL RISKS

While the dataset has been rigorously cleaned, the following empirical risks must be recognized for Week 2 modeling:

1. **Zero-Inflation (Intermittent Demand):** The post-cleaning dataset remains heavily populated with zero-sales records (\$\text{Median} = 0\$). Standard linear regression may output non-integer or slightly negative values, necessitating a post-processing clipping step (\$\hat{y} = \max(0, \hat{y})\$) or specialized evaluation (Poisson/Tweedie considerations).
2. **Loss of High-Volume Promo Effects:** By trimming sales above 2 units, genuine mega-promotional surges (e.g., Black Friday bulk buying) are excluded from the baseline model. A secondary regime-switching indicator or separate high-demand model could be explored in future work.
3. **Sampling Scaling:** The 100,000 sample validated the pipeline architecture. In Week 2, the identical node parameters will scale across the full 11.1 million transaction database.

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7. DELIVERABLE STATUS FOR WEEK 1 (PERSON 2 HAND-OFF)

- [x] **Relational SQL Ingestion:** Verified via `MySQL Connector` and `DB Query Reader`.
- [x] **Data Quality Assessment:** Formally audited across Completeness, Uniqueness, Validity, and Consistency.
- [x] **Automated Cleaning Pipeline:** Built and executed with `Missing Value` and `Numeric Outliers` nodes.
- [x] **Comparative Baseline Table:** Before/After metrics documented for Section 3 of the IEEE Final Project Report.
- [x] **Prepared for Week 2:** Cleaned feature space ready for `Partitioning` and `Linear Regression Learner` nodes.